

A decorative graphic on the left side of the slide consists of thin, grey lines that resemble a circuit board or a network diagram. These lines are vertical and horizontal, with small circles at various points, suggesting nodes or components. The lines are more dense on the left and become sparser towards the right.

INTRODUCTION TO SOFTWARE ENGINEERING

CE 325

Usama Arshad, PhD Cs

Usama.arshad@giki.edu.pk

Faculty of computers sciences and engineering (FCSE)

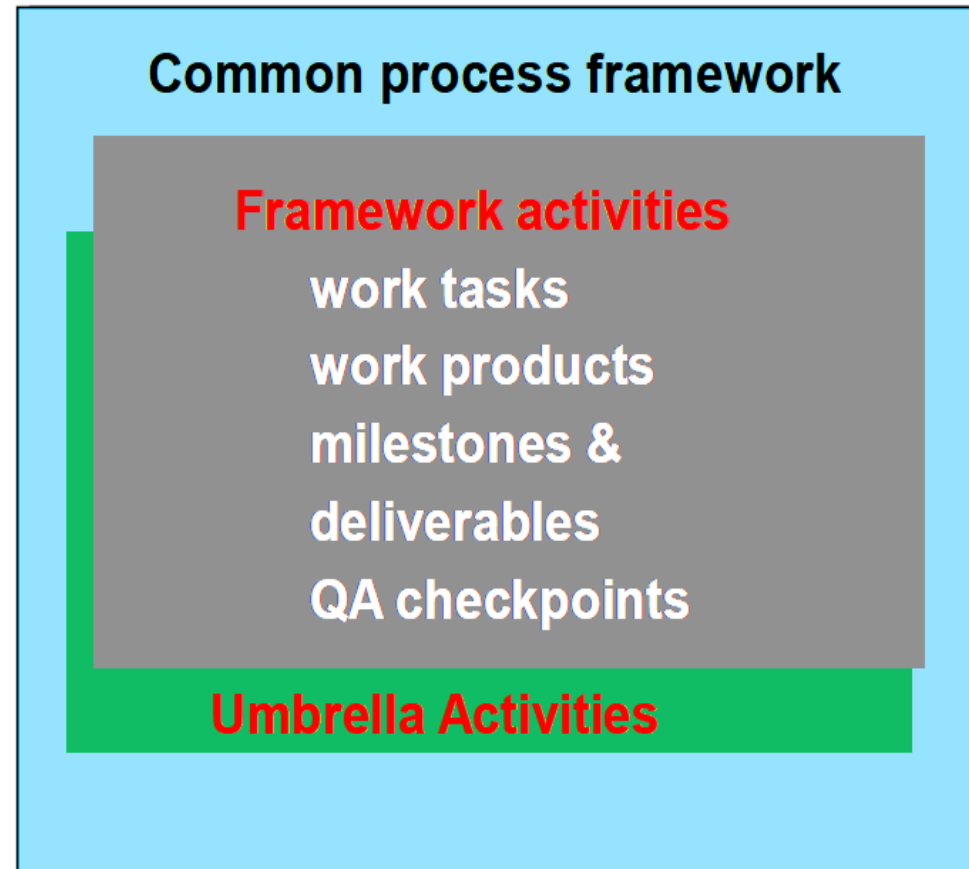


Ghulam Ishaq Khan Institute of Engineering Sciences and
technology (GIKI), Topi, Swabi, Pakistan

SOFTWARE PROCESS MODEL

DEVELOPED BY USAMA JANJUA

A Common Process Framework

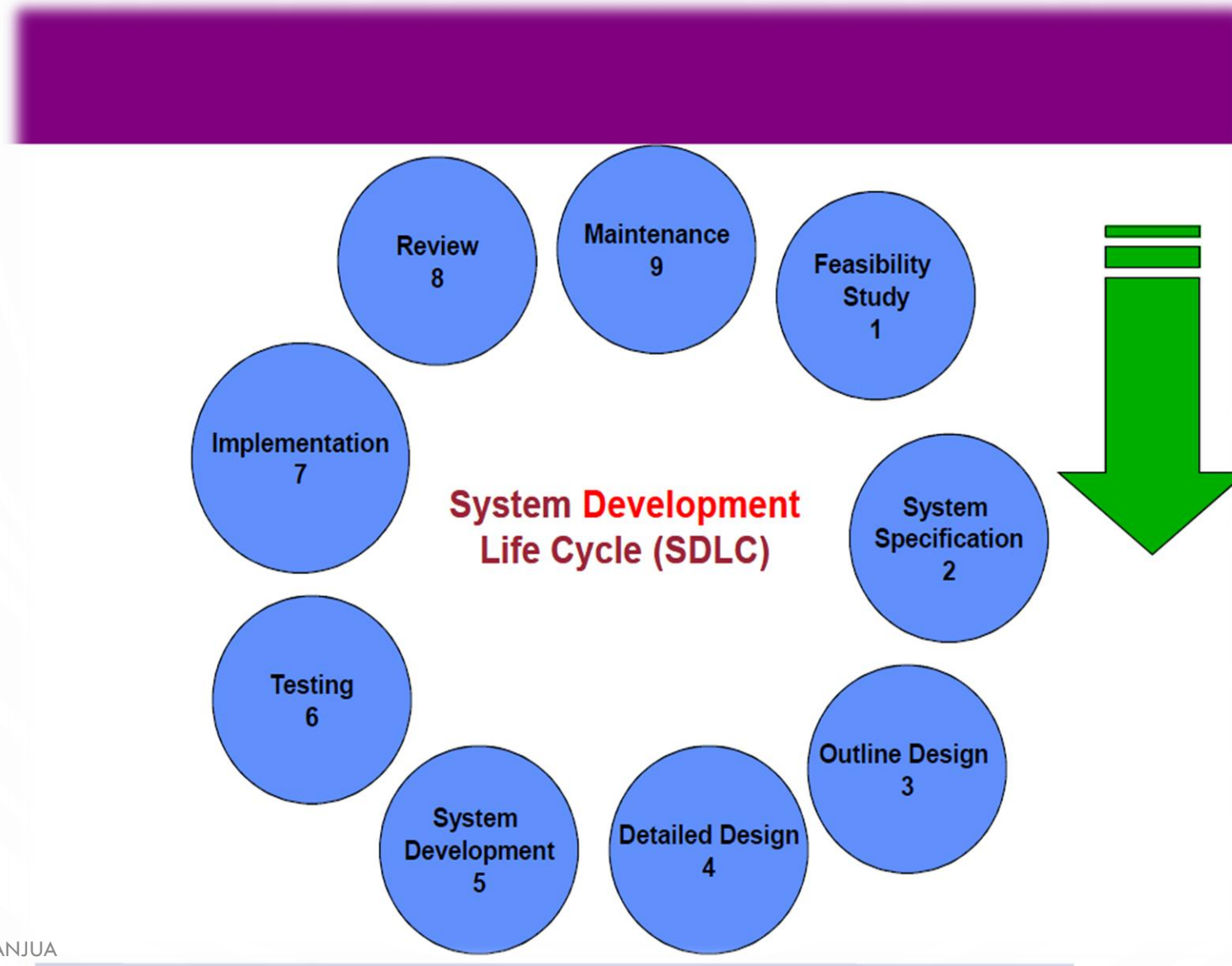


DEVELOPED BY USAMA JANJUA

Umbrella Activities

- **Software project management (SPM)**
- **Formal technical reviews (FTRs)**
- **Software quality assurance (SQA)**
- **Software configuration management (SCM)**
- **Document preparation and production**
- **Reusability management**
- **Measurement**
- **Risk management (RA&M)**

DEVELOPED BY USAMA JANJUA



DEVELOPED BY USAMA JANJUA

SOFTWARE PROCESS

- Process consists of activities/steps to be carried out in a particular order
- Software process deals with both technical and management issues
- Consists of different types of process
- Process for software development: produces software as end-result
 - multiple such processes may exist
 - a project follows a particular process

DEVELOPED BY USAMA JANJUA

PROCESS TYPES

- Process for managing the project
 - defines project planning and control
 - effort estimations made and schedule prepared
 - resources are provided
 - feedback taken for quality assurance
 - monitoring done.

DEVELOPED BY USAMA JANJUA

PROCESS TYPES

- Process for change and configuration mgmt.
 - Resolving requests for changes
 - Defining versions, their compositions
 - Release control
- Process for managing the above processes themselves
 - Improving the processes based on new techniques, tools, etc.
 - Standardizations and certifications (ISO, CMM)

DEVELOPED BY USAMA JANJUA

CHARACTERISTICS OF A GOOD PROCESS

- Should be precisely defined – no ambiguity about what is to be done, when, how, etc.
- It must be predictable – can be repeated in other projects with confidence about its outcome
 - Predictable with respect to effort, cost:

Project A: Web-based library applications done by 3 persons in 4 months

⇒ another project B (guest house bookings), similar in complexity should also take about 12 person months.

A GOOD PROCESS ...

- Facilitates early detection of and removal of defects
 - Defects add to project cost
 - Late detection/correction is costly
- It should facilitate monitoring and improvement
 - Based on feedback
 - Permit use of new tools, technologies
 - Permit measurements

DEVELOPED BY USAMA JANJUA

SOFTWARE PROCESS MODEL

The software engineers has five choices for the selection of software process models. The models are :

- Linear Sequential Model (LSM)
- The Prototype Model (PRM)
- The Rapid Application Development Model (RAD)
- The Incremental Model (INS)
- The Boehm Spiral Model (BMS)

In all models , core activities are Analysis, Design , Code, Test are common . However their execution differs from model to model.

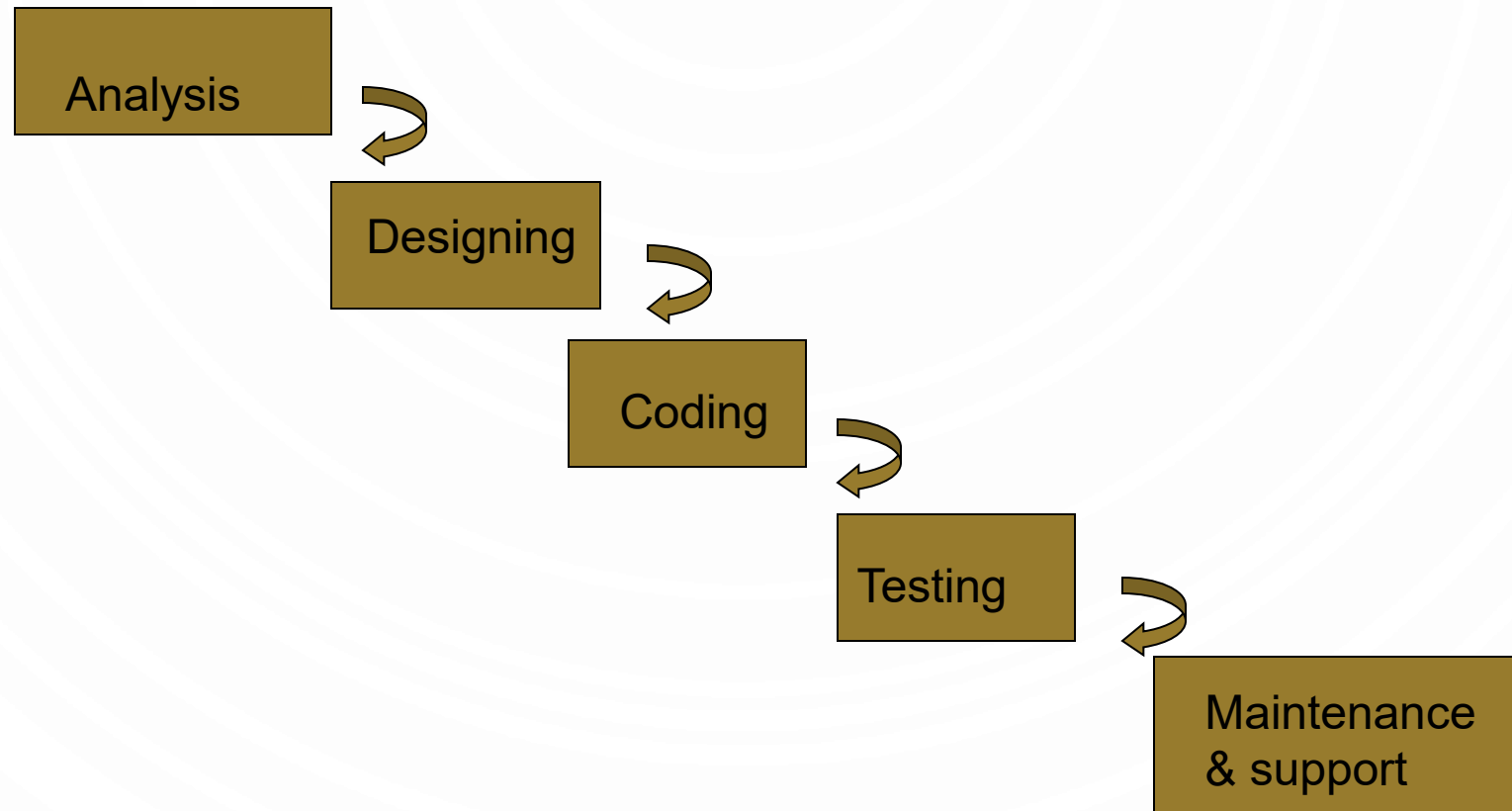
DEVELOPED BY USAMA JANJUA

THE LINEAR SEQUENTIAL MODEL (LSM)

- It is one of the earliest development models. The linear sequential model provides a systematic approach to software development. In this approach ,the process of software development is represented by a sequence of steps. The sequential phases are what make this model linear, simple and systematic in nature. Each phase must be completed before you can move to next phase. This model is also known as the Waterfall Model or classical life cycle .

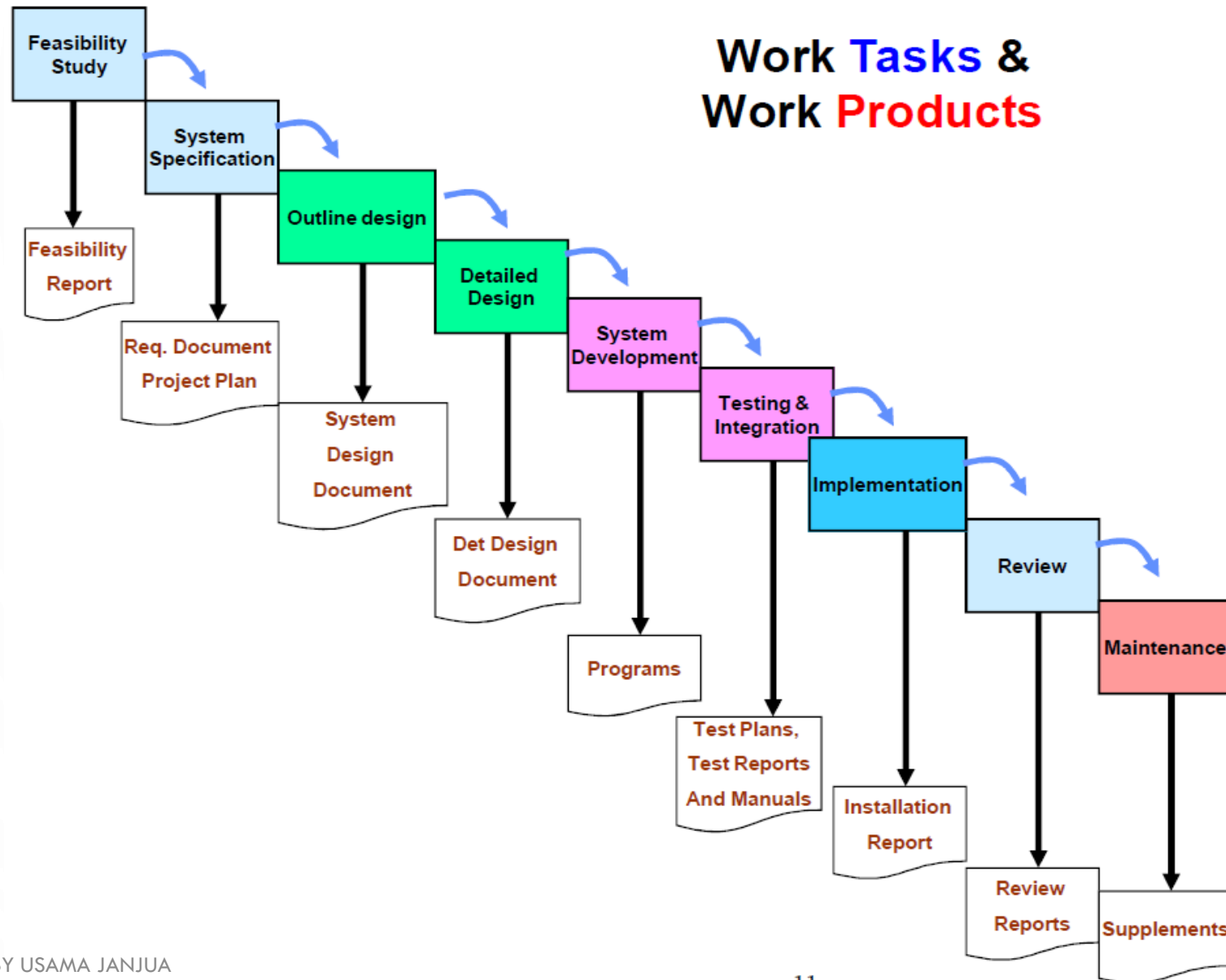
DEVELOPED BY USAMA JANJUA

PHASES OF LINEAR SEQUENTIAL MODEL



DEVELOPED BY USAMA JANJUA

Work Tasks & Work Products



DEVELOPED BY USAMA JANJUA

PHASES OF LINEAR SEQUENTIAL MODEL

The following are the phases of LSM:

- **System and Information Engineering:**

In this phase ,the requirements for all the elements are worked out.

System engineering denotes the requirements of system level and information engineering denotes the strategic requirements at business level.

- **Software requirement analysis:**

In this Phase , the requirements for the software are established through discussion with client and are then documented.

DEVELOPED BY USAMA JANJUA

PHASES OF LINEAR SEQUENTIAL MODEL

- **Designing:**

In this phase , the requirements are converted into design to be used in the next phase. The design includes various elements such as database structures, software architecture and procedures.

The design is first passes through the quality test and then documented.

- **Coding:**

In this phase the actual coding of the software is done. The design of the previous phase is converted into the code.

DEVELOPED BY USAMA JANJUA

PHASES OF LINEAR SEQUENTIAL MODEL

- **Testing:**

In this phase, the output generated is checked to ensure that it matches the requirements. The programs developed in the previous phase are checked for the logical and syntax errors.

- **Maintenance and Support:**

The software developed needs to maintenance and support. This refers to the changes as well as new requirements in the software after delivery.

ADVANTAGES OF LINEAR SEQUENTIAL MODEL

The Linear Sequential model offers the following advantages:

- It is easy to understand and implement.
- It prohibits skipping any phase in the sequence.
- It is ideal for small projects and when the requirements and goals of the project are well established in advance.

DEVELOPED BY USAMA JANJUA

DISADVANTAGES OF LINEAR SEQUENTIAL MODEL

The following are the disadvantages using Linear sequential model:

- In practical conditions ,software projects are seldom sequential. Iteration and overlapping often occur . This model is not suitable for such projects.
- The working version of the software is available to the customer after testing. Therefore, if there is any major error during the coding it will till end of the testing.
- Due to linear nature is any phase is not completed , the software analyst and developers cannot proceed further.

DEVELOPED BY USAMA JANJUA

Total Feedback



DEVELOPED BY USAMA JANJUA

Phone Directory: Requirements

Phone Directory:

- Interactive Program containing collection of names and phone-numbers
- Insert new entries
- Retrieve entries
- Change Entries

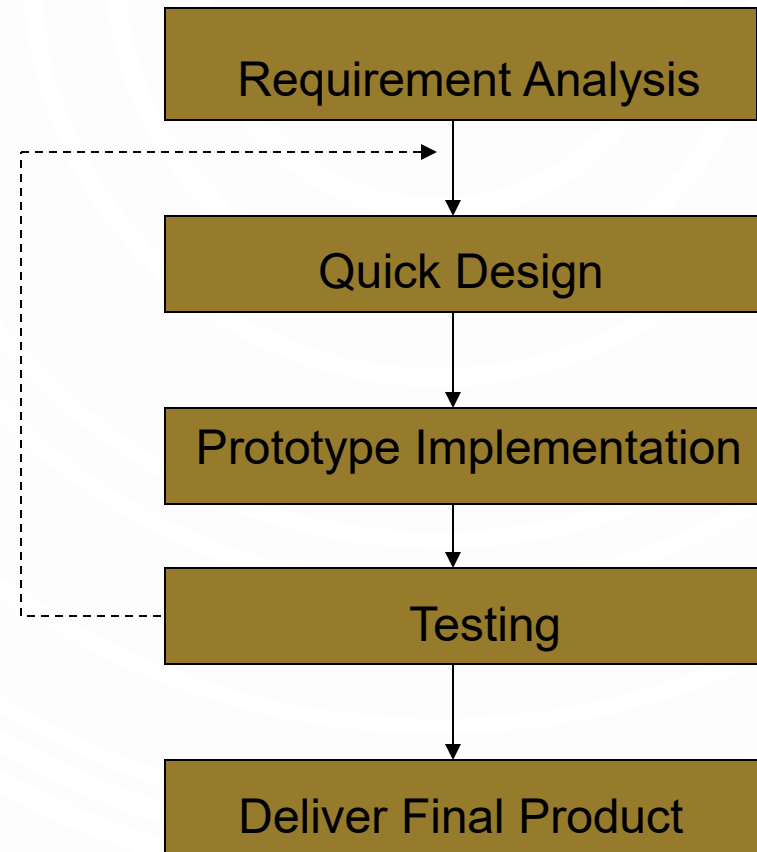
DEVELOPED BY USAMA JANJUA

PROTOTYPING MODEL

- In this model the developer and client interact to establish the requirements of the software.
- Define the broad set of objectives.
- This is followed by the quick design, in which the visible elements of the software, the input and the output are designed.
- The quick design stresses the client's view of the software.
- The final product of the design is a prototype.
- The client evaluates the prototype and provides its recommendations and suggestions to the analyst.
- The process continues in an iterative manner until all the user requirements are met.

DEVELOPED BY USAMA JANJUA

PHASES OF PROTOTYPING MODEL



DEVELOPED BY USAMA JANJUA

PROTOTYPING MODEL

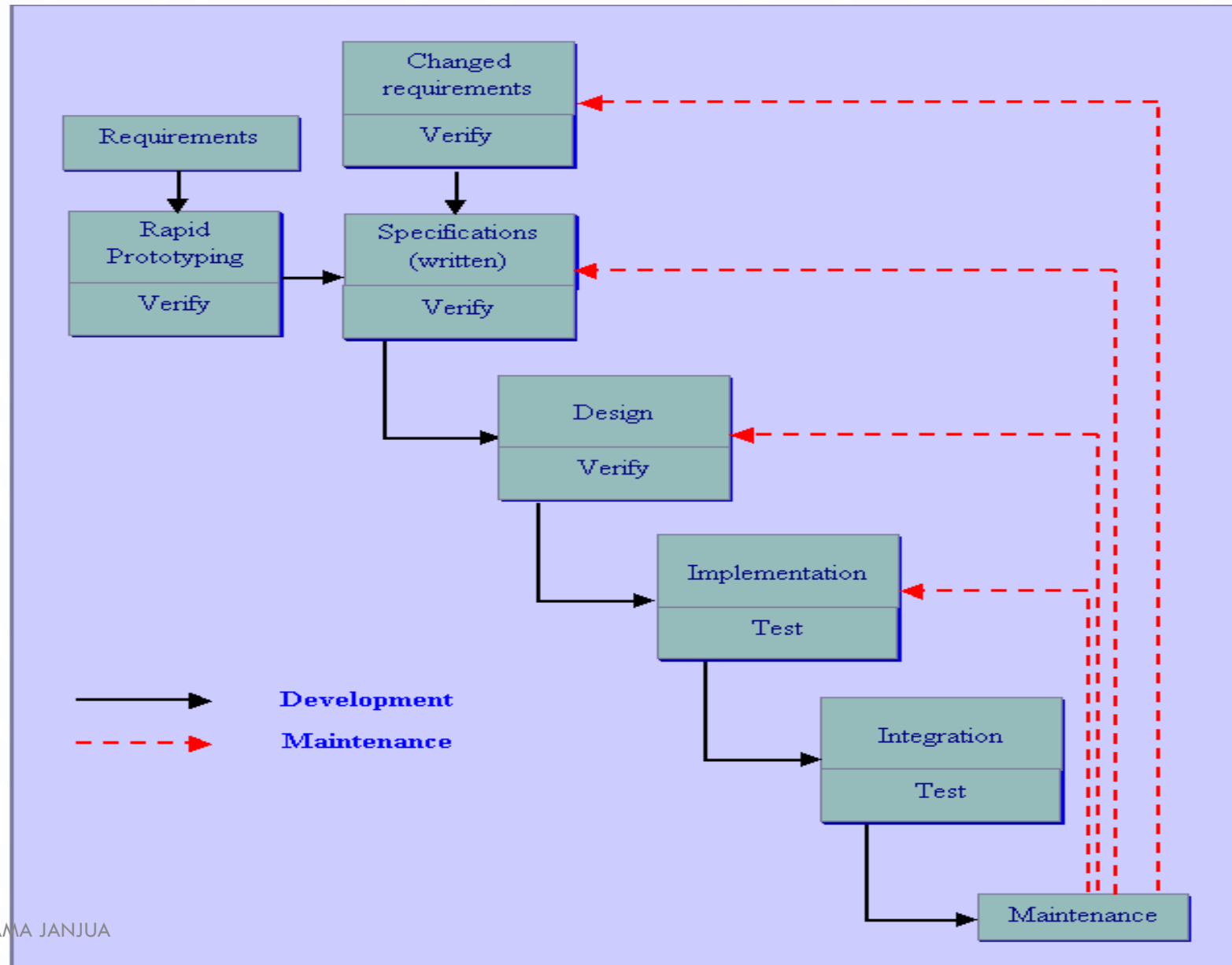
- The following sections examines two (2) version of the prototyping model:
- **Version I**: Prototyping is used as a requirements technique.
- **Version II**: Prototype is used as the specifications or a major part thereof.

DEVELOPED BY USAMA JANJUA

VERSION-I

- This approach, as illustrated in fig on next slide, uses the prototype as a means of quickly determining the needs of the client; it is discarded once the specifications have been agreed on. The emphasis of the prototype is on representing those aspects of the software that will be visible to the client/user (e.g. input approaches and output formats). Thus it does not matter if the prototype hardly works.
- Note that if the first version of the prototype does not meet the client's needs, then it must be rapidly converted into a second version.

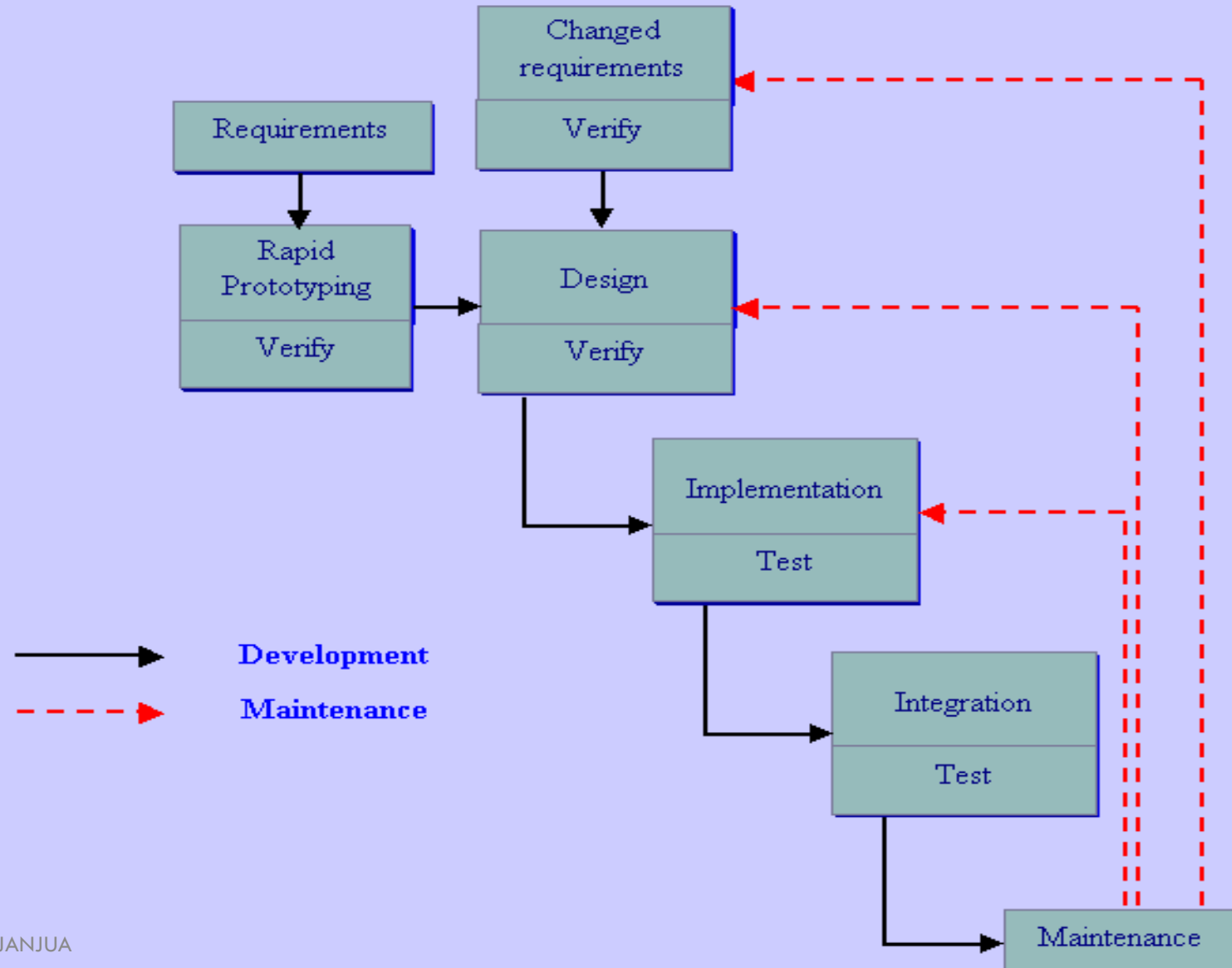
DEVELOPED BY USAMA JANJUA



VERSION-II

- In this approach, as illustrated in Fig, the prototype is actually used as the specifications for the design phase. This advantage of this approach is speed and accuracy, as not time is spent on drawing up written specifications. The inherent difficulties associated with that phase (i.e. incompleteness, contradictions and ambiguities) are then avoided

DEVELOPED BY USAMA JANJUA



DEVELOPED BY USAMA JANJUA

ADVANTAGES OF PROTOTYPING MODEL

The following are the advantages of Prototyping model:

- Due the interaction between the client and developer right from the beginning , the objectives and requirements of the software is well established.
- Suitable for the projects when client has not clear idea about his requirements.
- The client can provide its input during development of the prototype.
- The prototype serves as an aid for the development of the final product.

DEVELOPED BY USAMA JANJUA

DISADVANTAGES OF PROTOTYPING MODEL

The prototyping model has the following disadvantages.

- The quality of the software development is compromised in the rush to present a working version of the software to the client.
- The client look at the working version of the product at the outset and expect the final version of the product to be deliver immediately. This cause additional pressure over the developers to adopt shortcut in order to meet the final product deadline.
- It becomes difficult for the developer to convince the client as why the prototype has to be discarded.

DEVELOPED BY USAMA JANJUA

Problems to be tackled by Prototyping

- ✓ **Customer's misconception** (req-completion/communication, involvement, expectations, automation)
- ✓ **Developer's wrong habits** (Not strict to req, misunderstanding, assumptions/imaginations)
- ✓ **Rules of the game are not defined at the beginning**

DEVELOPED BY USAMA JANJUA

RAPID APPLICATION DEVELOPMENT:

- RAD is a high speed version of linear sequential model. It is characterized by a very short development life cycle, in which the objective is to accelerate the development.
- The RAD model follows a component based approach.
- In this approach individual components developed by different people are assembled to develop a large software system.

DEVELOPED BY USAMA JANJUA

PHASES OF RAPID APPLICATION DEVELOPMENT:

The RAD model consist of the following phases.

- **Business Modeling:**

In this phase, define the flow of information within the organization, so that it covers all the functions. This helps in clearly understand the nature, type ,source and process of information.

- **Data Modeling:**

In this phase, convert the component of the information flow into a set of data objects. Each object is referred as an *Entity*.

PHASES OF RAPID APPLICATION DEVELOPMENT:

- **Process Modeling:**

In this phase, the data objects defined in the previous phase are used to depict the flow of information. In addition adding, deleting, modifying and retrieving the data objects are included in process modeling.

- **Application Designing:**

In this phase, the generation of the application and coding take place. Using fourth generation programming languages or 4 GL tools is the preferred choice for the software developers.

- **Testing:**

In this phase, test the new program components.

DEVELOPED BY USAMA JANJUA

ADVANTAGES OF USING RAD MODEL:

The RAD has following advantages:

- Due to emphasis on rapid development , it results in the delivery of fully functional project in short time period.
- It encourages the development of program component reusable.

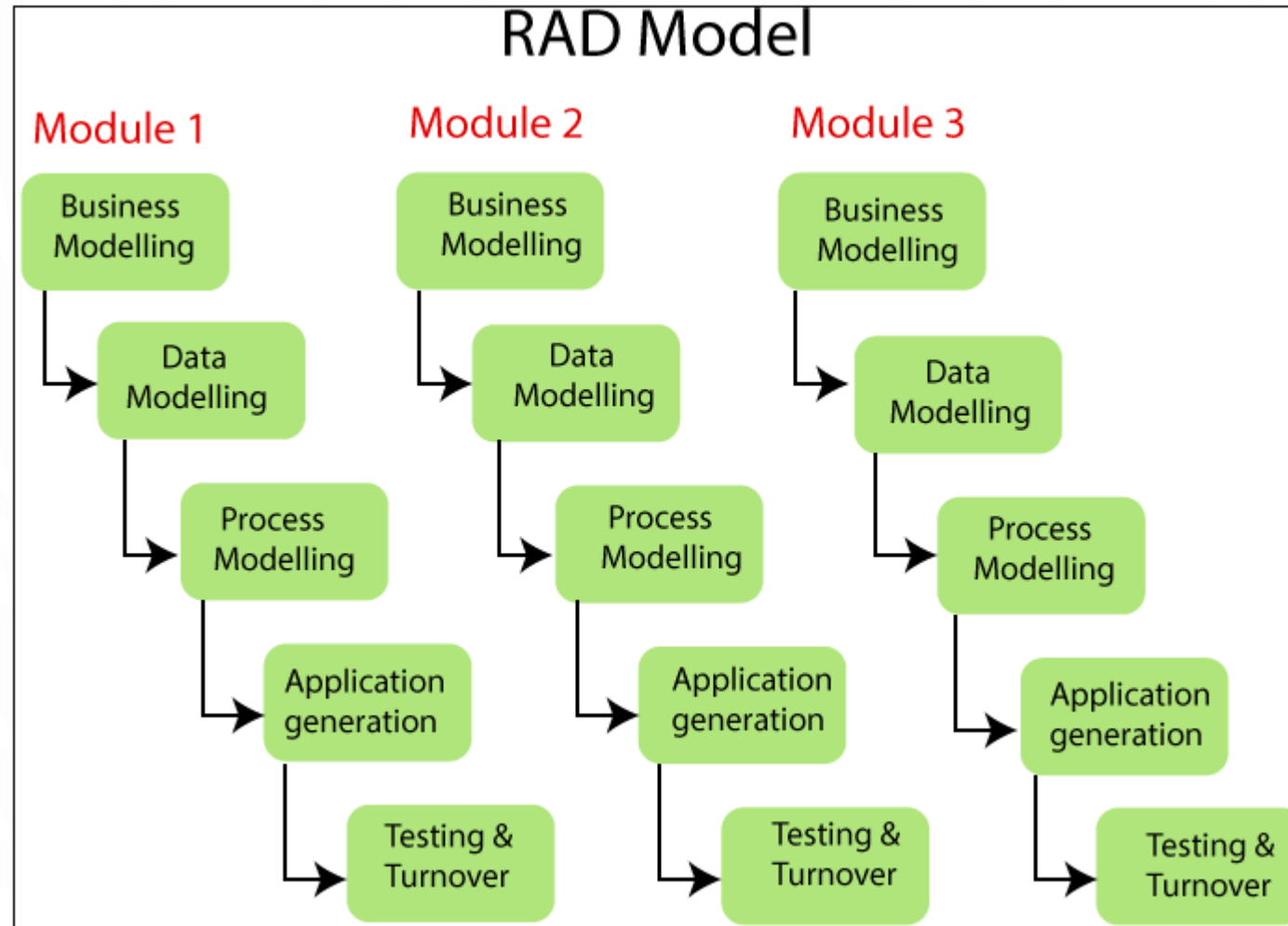
DEVELOPED BY USAMA JANJUA

DISADVANTAGES OF USING RAD MODEL:

The RAD model has following disadvantages :

- It requires dedication and commitment on the part of the developers as well as the client to meet the deadline. If either party is indifferent in needs of other, the project will run into serious problem.
- Its application area is restricted to system that are modular and reusable in nature.
- It is not suitable for the applications that have a high degree of technical risk.
- It is not suitable for the large projects because they require more manpower for creating multiple RAD groups.

DEVELOPED BY USAMA JANJUA



DEVELOPED BY USAMA JANJUA

INCREMENTAL MODEL (INM)

- The incremental model is the combination of the features of linear sequential model and the iterative approach of the prototyping model.
- The software is developed and delivered in small increments and the linear sequential model is applied to each increment.
- In an incremental model ,the prototyping methodology is applied to each process flow of each increment.
- In the case of the incremental model , the first increment that is delivered is the core product.
- The core product addresses the primary needs of the final product.
- It is evaluated and reviewed by the client.

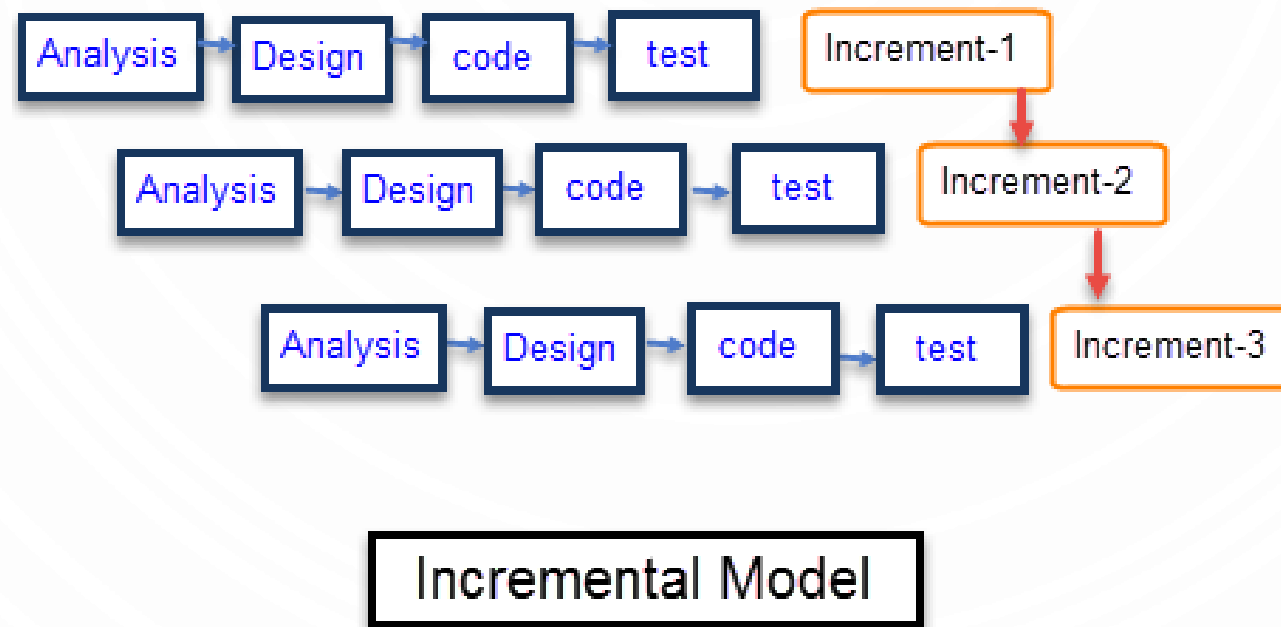
DEVELOPED BY USAMA JANJUA

INCREMENTAL MODEL (INM)

- At times, it is even used on a test basis. Based on client feed back for the core product developers prepare a plan for the next increment.
- New features and functionality are also taken into consideration.
- The above process is applied to each increment that is delivered.

DEVELOPED BY USAMA JANJUA

INCREMENTAL MODEL (INM)



DEVELOPED BY USAMA JANJUA

Advantages of using the Incremental Model (INM)

This model has following advantages:

- Compared to RAD, it requires less human resources, especially for the first few increments.
- It guarantees early delivery of the final products each increment leads to the development of the software.
- If any resource staff is unavailable , it does not affect or delay the project.
- The incremental model is therefore ideal for those projects in which sufficient manpower is not available to meet difficult project dealines.

DEVELOPED BY USAMA JANJUA

DISADVANTAGES OF USING INCREMENTAL MODEL (INM)

- Similar to the software prototyping model , software quality is sometimes compromised in the rush to finish the software earlier than planned.

DEVELOPED BY USAMA JANJUA

THE SPIRAL MODEL

- The spiral model developed by Boehm combines the philosophy of INM, RAD and LSM models with the use of prototyping.
- The spiral model is recommended where the requirements and solution call for developing full-fledge , large, complicated system with lots of features and facilities from the scratch.
- It is used when experimenting on technology , trying out new skills and when the user is not able to offer requirements in clear terms.
- It emphasis at the quick development of the software, which is released in increments.

DEVELOPED BY USAMA JANJUA

THE SPIRAL MODEL

This model consists of a number of activities called task regions.

The number of task regions varies from three to six.

A spiral model consist of the following task regions.

- Communication
- Planning
- Risk analysis
- Engineering
- Construction and release
- Evaluation

DEVELOPED BY USAMA JANJUA



ADVANTAGES OF USING SPIRAL MODEL

The spiral model has the following advantages :

- This model is more in tune with large real-life project development.
- Prototyping can be applied at any level in evaluation process.
- This helps to reduce technical risk.
- It is suitable for application that can be use an object oriented approach to develop software.

DEVELOPED BY USAMA JANJUA

DISADVANTAGES OF USING SPIRAL MODEL

The spiral model has the following disadvantages:

- It can cause problems in negotiating a development contract with the client.
- It requires considerable experience in risk management for the project to be successful.
- It requires a lot of patience and time in years before this model's effectiveness can be assessed accurately.

DEVELOPED BY USAMA JANJUA